

The Utility Problem

Accurate load forecasting underpins almost every critical decision a utility makes — from dispatch and procurement to capital investment and grid reliability.

Resource Planning & Energy Management - electricity cannot be stored at grid scale, and generation must be balanced with consumption in real time. Accurate load forecasting allows utilities to plan better for future energy generation and distribution needs. This ensures sufficient generation capacity to meet peak demand and prevent blackouts or brownouts. Better forecasting also informs decisions about when to generate, procure, or dispatch stored energy — reducing costs and minimizing curtailment.

Asset Management - load forecasts are essential for capacity planning: determining whether existing infrastructure can absorb new load growth before thermal limits or voltage constraints are breached. Given the long lead times for grid reinforcement — substation upgrades, new line construction, and associated permitting can take years — utilities must anticipate capacity constraints well in advance.

Energy Procurement - load forecasts inform wholesale market participation, helping utilities optimize the timing and volume of energy purchases and sales to minimize procurement costs.

Maintenance Scheduling - load forecasts allow utilities to schedule maintenance windows during low-demand periods, reducing the risk of unplanned outages and improving overall system reliability.

Regulatory Compliance - load forecasts support adherence to grid codes and balancing obligations, including mandatory reporting to transmission system operators (TSOs) and regional reliability organizations.

Customer Engagement - by anticipating high-demand periods, utilities can proactively communicate potential service disruptions and design demand response programs or load-shifting incentives to reduce peak load.

Across all these dimensions, accurate short-term and long-term load forecasting is fundamental to utility operations. The growing penetration of distributed energy resources — rooftop solar, EVs, and battery storage — is further complicating load patterns, making data-driven forecasting tools more critical than ever.

The Data Solutions and Methods

Electrical load forecasting can be approached through several established methods:

- Time-series analysis - this method uses historical data to make predictions about future electrical loads. It can be further divided into methods such as moving averages, exponential smoothing, and ARIMA models.
- Artificial neural networks (ANNs) - this machine learning method models the non-linear relationship between input variables — such as temperature, time of day, and day of week — and electrical load.
- Support vector regression (SVR) - a supervised machine learning method that trains on historical input-output pairs to predict future load. SVR performs well with high-dimensional data and is robust to outliers.
- Econometric models - this method uses economic and demographic data to predict future electrical loads.
- Hybrid models - this approach combines two or more of the above methods to improve forecast accuracy, often outperforming any single method alone.

The choice of method depends on the specific requirements and available data. The Awesense Platform supports all of the methods above, providing analysts with the time-series infrastructure and data access functions needed to implement any forecasting approach.

The example below demonstrates a simple time-series approach using a Simple Moving Average (SMA). The main advantage of moving averages is that it is simple to implement and understand, but they can be less accurate when load data exhibits high volatility or when the window size is not well-calibrated. This method applies an SMA at the distribution feeder level.¹ (SMA) on the resolution of a feeder. The SMA method calculates the average of the past n observations. The number of past observations used in the calculation is called the window size. For example, if the window size is 3, the SMA would be the average of the past three observations.

To obtain the load data for a feeder, analysts using the Awesense Platform have two options:

1. Getting the time-series data directly from the top-of-the-feeder element (usually breaker) *ts_data_source_select()*.
2. If the time-series measurements are unavailable, analysts can infer the data by aggregating the downstream load. This can be accomplished by iterating through top-of-the-feeder elements and fetching the elements “with loads” (usually meter elements). This can be achieved by *grid_get_downstream()* or *grid_get_connected()* functions. After the meters are fetched, *ts_data_source_select()* will yield a consumption time-series, which can be aggregated and the feeder load can be calculated.

With historical feeder level load time-series, analysts can now employ the SMA method to calculate the load forecasting. As noted above, any of the forecasting methods described can be

¹ Other moving averages are possible as well, specifically Weighted Moving Average (WMA) or Exponentially Weighted Moving Average (EWMA). Both put different emphases on more recent data and how the most recent data contribute to the overall result. The choice of moving average technique and window size depends on the specific requirements and characteristics of the data.

Load Forecasting

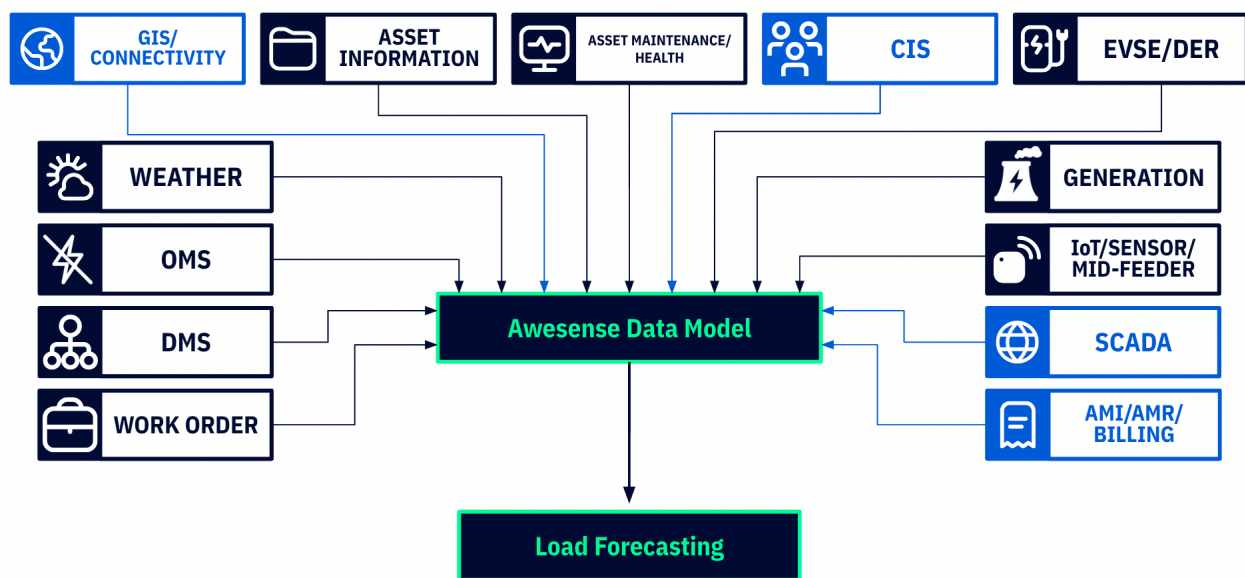
[UC43] - Awesense Platform Use Case



applied here. A natural extension is to combine SMA with econometric models, incorporating location-specific variables such as regional load growth trends or EV adoption rates.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. Please contact [Awesense](#) if you are interested in the implementation of this use case.